

Name: Matt Egeler

80

CHE 321; Spring - 2020; Quiz - 3

100 Points

1. (25 points) Three gas phase reactions have the same reaction rate dependence to the concentration of the reactant.

- a. ~~$A \rightarrow B$~~
b. $A \rightarrow C$
c. $2A \rightarrow D$

Answer: C

Which of these reactions will exhibit the highest conversion in a PFR? Explain.

The rate of conversion in a PFR depends on the concentration of A, so if there is constantly more A there will be higher conversion. As the reaction goes forward the volumetric flow rate decreases which increases the concentration of A.

2. (25 points) Why does the conversion for a gas-phase reaction in a PBR depend on the pressure?

The pressure controls how long each molecule spends in the reactor, so the pressure controls the conversion.

as pressure drop is significant for gas-phase reaction in PBR

3. (25 points) Two reactions (one gas phase and one liquid phase) have the same reaction rate dependence to the concentration of the reactant.

a. $A \rightarrow 1/3 B$ (Liquid Phase)

b. $A \rightarrow 1/3 B$ (Gas Phase)

Which of these reactions will exhibit the highest conversion in a PFR? Explain.

The volumetric flow rate will keep decreasing which increases the concentration of A which increases the conversion.

4. (25 points) A reaction rate equation is given by $r_A' = k' P_A$. What are the units of k' ?

$$r_A' [=]$$

$$k' [=] \frac{\text{Concentration}}{\text{Pressure} \cdot \text{Time}} \cdot \text{mass}$$

$$r_A' = \frac{[\text{mol}]}{[\text{kg}][\text{s}]} = k' \cdot P_A = k' [\text{atm}]$$

$$\Rightarrow k' = \frac{[\text{mol}]}{[\text{kg}][\text{s}][\text{atm}]}$$